Orbital Contact Analyzer — User Manual

Version: v1.3.0 **Scope:** Single-file, offline HTML app for simulating two Earth satellites and up to two ground stations. The application uses a J2 perturbation model, renders 2D/3D views, visualizes sensor footprints and local horizons, and computes both satellite-to-satellite line-of-sight (LOS) and satellite-to-ground-station access intervals.

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1) Overview

This app propagates two satellites using a model that includes J2 secular perturbations, converts positions between ECI and ECEF frames, and visualizes:

- **2D Map:** Equirectangular projection with ground tracks, current markers, sensor footprints (FOV), local horizons, and communication links.
- **3D Globe:** WebGL-based view with full camera control, physically correct inertial and Earth-fixed views, and a full-screen "Focus Mode."
- **Scenario Loader:** A quick-configuration menu to instantly load common orbital regimes (LEO, GEO, Molniya, etc.).
- **Data Tables:** Lists for Inter-Satellite LOS and Ground Station Access intervals, with a single-click CSV export.

Key design choices:

- **J2 Perturbed propagation:** Includes secular effects on RAAN (Ω) and Argument of Perigee (ω).
- **Spherical Earth model:** Used for all ground geometry, occlusion checks, and footprint calculations.
- **Global Epoch:** The simulation timeline is anchored to a single, user-defined UTC epoch.

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2) System Requirements

- A modern desktop browser with **WebGL enabled** (Chrome, Edge, Firefox, Safari).
- 8+ GB RAM recommended for long timelines or small time steps.
- **Runs fully offline;** no external network access is required.

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3) Concepts in Brief

- **Orbital Elements (per satellite):**

- a (km), e, i (deg), Ω (RAAN), ω (ArgP), M (Mean Anomaly).
    - **Min. elevation ϵ (deg):** Defines the sensor footprint (Field of View).
- **Ground Stations:** Up to two points on the Earth's surface, each defined by Latitude, Longitude, and a Min. Elevation Mask (deg) used to calculate access. Ground Station 1 (GS1) is shown as a white triangle; Ground Station 2 (GS2) is shown as a green triangle.
- **J2 Perturbations:** The application models the primary secular (long-term) effects of Earth's equatorial bulge, which causes the orbital plane (Ω) and the orbit's orientation within that plane (ω) to precess over time.
- **Frames:**
    - **ECI** (Earth-Centered Inertial) for orbital dynamics.
    - **ECEF** (Earth-Centered Earth-Fixed) for surface overlays.
- **Inter-Satellite LOS:** A direct line-of-sight exists if the straight line connecting the two satellites is not obstructed by the spherical Earth.

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4) Quick Start

1. **Open the App:** Open the `index.html` file in your browser.
2. **Load a Scenario (Recommended):** At the top of the **Simulation Controls** panel, use the **Load Scenario** dropdown to instantly configure the simulation.
    - *Select "LEO Formation":* To see two satellites in low Earth orbit passing over a ground station.
    - *Select "Sun-Synchronous":* To demonstrate J2 perturbation effects over a 24-hour period.
    - *Select "Molniya":* To view high-eccentricity orbits with long dwell times over the northern hemisphere.
3. **Animate:** Click the green ▸ **Run** button to start the time evolution.
    - Use the **Scrub Bar** (bottom of the view) to manually drag time forward or backward.
    - *Note:* If you manually change any input number (e.g., Altitude), the scenario dropdown will automatically switch to **"Custom"**, preserving your edits.
4. **Visualize:**
    - Use the **Projection** dropdown (top right) to switch between the **2D Map** and **3D Globe**.
    - Click **Focus Mode** (↗) to expand the visualization to fill your screen.
5. **Analyze Data:** Scroll down to view the **Line-of-Sight** and **Ground Station Access** tables. Click **Export Contact Times (CSV)** to download a spreadsheet of all computed windows.

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5) Controls Reference

5.1 Simulation Controls (Top-Left)

- **Load Scenario:** A preset menu that configures the entire application state (Satellites, Ground Stations, and Time settings) for specific astrodynamics demonstrations.

- **Epoch (UTC, ISO-8601):** Global start time (e.g., `2025-11-18T12:00:00Z`).
- **Step Δt (s):** Simulation step size. Smaller is more precise but computationally heavier.
- **Duration (min):** Total span simulated from the epoch.
- **Playback speed:** Visual playback rate when running (steps per animation frame).
- **Buttons:**
  - ▶ **Run / Pause:** Start or stop time stepping.
  - ↺ **Reset:** Rewind to the epoch.
  - **Recompute tracks:** Re-runs the simulation with current inputs (useful if you change an orbit manually).
  - **Export Contact Times (CSV):** Saves all calculated data to a file.

5.2 Satellite Panels

- **Elements ($a,e,i,\ldots$):** Standard Keplerian inputs.
- **Min. elevation ϵ:** Defines the size of the dashed FOV circle.
- **Visibility Toggles:** Show/hide the Satellite marker, Track, FOV circle, or Horizon circle.

5.3 Ground Station Panels

The application supports two independent ground stations. Ground Station 1 (GS1) is enabled by default; Ground Station 2 (GS2) is disabled by default and can be enabled when needed.

Each ground station panel contains:

- **Show Ground Station:** Toggles visibility and enables access calculations for that station.
- **Lat/Lon:** Station coordinates.
- **Min Elev. Mask:** The minimum elevation angle (above the local horizon) required for a satellite to be considered "in view" from that station.

On the 2D map and 3D globe, GS1 appears as a **white** triangle and GS2 appears as a **green** triangle. When a satellite has access to a ground station, a dashed link line is drawn between them in the satellite's color.

The **Ground Station Access Intervals** table (below the map) merges intervals from both stations in chronological order. Each row indicates which station (GS1 or GS2) and which satellite are involved. The footer displays separate real-time badges for each station.

The **CSV export** labels intervals as `ground-station-1` or `ground-station-2` in the type column.

5.4 View Controls (Right-Side)

- **Projection:** Toggle 2D Map vs. 3D Globe.
- **Focus Mode (↗):** Expands view to full window.
- **3D Controls:**
  - **Reset Camera:** Re-centers the view.
  - **Focus:** Automatically follows Sat 1 or Sat 2.
  - **Inertial View:**

- *ON:* Orbital tracks are fixed in space; Earth rotates underneath (Physically correct).
- *OFF:* Camera rotates with the Earth (ECEF view).

- **Show Elements:** Toggles the HUD showing live orbital elements (Ω,ω).

5.5 Keyboard Shortcuts

The following keyboard shortcuts are available for faster navigation. Shortcuts are disabled when typing in input fields.

Key	Action
`Space`	Play/Pause simulation
`R`	Reset to start
`←` / `→`	Step backward/forward (auto-pauses)
`V`	Toggle 2D/3D view
`I`	Toggle Inertial view (3D only)
`0`	Free camera (no focus)
`1`	Focus on Satellite 1
`2`	Focus on Satellite 2
`E`	Export CSV
`T`	Open TLE import for Satellite 1
`Shift+T`	Open TLE import for Satellite 2

5.6 TLE Import

Each satellite panel includes a collapsible **"Import from TLE"** section that allows you to populate orbital elements from a standard Two-Line Element set.

To use TLE Import:

1. Click **"▸ Import from TLE"** at the bottom of a satellite panel (or press `T` for Sat 1, `Shift+T` for Sat 2).
2. Paste a TLE (2 or 3 lines) into the text area.
3. Optionally check **"Set simulation epoch to TLE epoch"** to sync the simulation start time.
4. Click **"Apply TLE"** to populate the orbital elements.

Supported TLE format:

```
ISS (ZARYA)             ← Optional name line (ignored)
1 25544U 98067A   24001.50000000  .00016717  00000-0  10270-3 0  9025
2 25544  51.6400 247.4627 0006703 130.5360 229.6106 15.49815571423456
```

What gets extracted:

- Inclination, RAAN, Eccentricity, Argument of Perigee, Mean Anomaly (converted to True Anomaly)

- Mean Motion (converted to Semi-major Axis)
- Epoch (year and day-of-year converted to ISO-8601 UTC)

Understanding TLE Epochs

What is a TLE epoch?

Think of a TLE as a photograph of a satellite's orbit taken at a specific moment. That moment is called the "epoch." The TLE tells you exactly where the satellite was and how it was moving when the photograph was taken. Just like a photograph, a TLE captures a single instant in time.

The mismatch problem

Imagine you have two photographs of friends taken on different days. If you try to place them in the same scene, something will look wrong because they weren't actually together at the same time.

The same thing happens with TLEs. Suppose:

- Satellite 1's TLE is from **January 15th**
- Satellite 2's TLE is from **January 20th**
- Your simulation starts on **January 15th**

Satellite 1 will be shown correctly because its TLE matches the simulation date. But Satellite 2 will appear where it *would have been* on January 20th, not where it *actually was* on January 15th. Over five days, a LEO satellite completes roughly 75 orbits, so the position error can be substantial.

Why positions drift over time

Satellites don't stay still. Between any two dates:

- The satellite completes many orbits, changing its position along the orbital path
- Earth's equatorial bulge causes the orbital plane to slowly rotate (RAAN drift)
- The orientation of the orbit's ellipse also shifts (argument of perigee drift)

These effects accumulate. A TLE from last week won't accurately describe where a satellite is today.

Normalized vs. raw TLEs

You may notice that some TLEs always show satellites starting at the equator. This happens because certain data providers "normalize" their TLEs so the epoch aligns with the satellite's ascending node (the moment it crosses the equator heading north). This is a processing choice, not a physical requirement.

Raw, unprocessed TLEs from sources like CelesTrak or Space-Track will show satellites at whatever position they occupied when the TLE was generated. Both types are valid; they're just different conventions.

Practical recommendations

- When comparing two satellites, use TLEs with epochs within 1 day of each other
- Enable "Set simulation epoch to TLE epoch" for at least one satellite to anchor the simulation to real orbital data

- If you need both satellites positioned accurately, find TLEs from the same source generated on the same day
- For educational or demonstration purposes, exact epoch matching is less critical

Caution: Dual epoch sync

If you enable "Set simulation epoch to TLE epoch" for *both* satellites and their TLE epochs differ, the behavior is "last write wins":

1. Import TLE for Satellite 1 with epoch sync → simulation epoch becomes Satellite 1's TLE epoch
2. Import TLE for Satellite 2 with epoch sync → simulation epoch is *overwritten* with Satellite 2's TLE epoch

This means Satellite 1's elements (which were accurate at its TLE epoch) are now being propagated from a different epoch, which could introduce error. To avoid this, either use TLEs with matching epochs, or enable epoch sync for only one satellite.

The app will warn you if a TLE's epoch differs from the simulation epoch by more than 1 day. After import, expand the TLE section to view the epoch for reference.

Note: The TLE parser validates that the resulting orbit has perigee above Earth's surface. Invalid TLEs will display an error message.

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6) How The Physics Works

- **Orbit Propagation:** Two-body Keplerian motion with J2 secular perturbations applied to RAAN (Ω) and Argument of Perigee (ω) at every time step.
- **ECI ↔ ECEF:** Standard transformation using Greenwich Mean Sidereal Time (GMST).
- **Horizon/FOV:** Calculated assuming a **Spherical Earth** radius ($R_E = 6378.137$ km).
- **LOS Test:** Geometric intersection test between the satellite-to-satellite segment and the Earth sphere.

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7) Typical Workflows

A) Visualize a Sun-Synchronous Orbit

1. In **Simulation Controls**, select **Load Scenario: Sun-Synchronous (J2 Demo)**.
2. Switch to **3D Globe**.
3. Enable **Inertial View** and check **Show Elements**.
4. Scrub the timeline. You will see the orbital plane (Ω) slowly precessing, validating the J2 physics engine.

B) Follow a Satellite in Inertial Space

1. Select **Load Scenario: LEO Formation**.
2. Switch to **3D Globe** and enable **Inertial View**.
3. Set **Focus: Sat 1**.
4. The camera will now "chase" the satellite's position in space while the Earth rotates beneath it.

C) Analyze High-Latitude Access (Molniya)

1. Select **Load Scenario: Molniya**.
2. Observe the 2D Map. Notice how the satellite spends the majority of its time (the "apogee dwell") over the northern hemisphere, maximizing contact with the high-latitude Ground Station.

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8) Simulation Fidelity & Limitations

This tool is designed for educational demonstration and first-order analysis.

- **J2 Dynamics:** The model includes J2 secular perturbations but **does not** include atmospheric drag, solar radiation pressure, or third-body gravity (Moon/Sun).
- **Spherical Ground Model:** All ground calculations assume a perfect sphere.
  - *Implication:* Ground tracks and footprints may have slight deviations compared to a WGS-84 ellipsoid model, particularly at high latitudes.
  - *Note:* While visually sufficient for demos, mission-critical analysis should use high-fidelity software (STK, GMAT).

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9) Troubleshooting

- **Blank Screen / No Tracks:** Ensure your browser supports WebGL. Check that the Epoch is in a valid ISO-8601 format (e.g., `2025-01-01T12:00:00Z`).
- **Can't Rotate Globe:** Check if **Focus** is set to a satellite. Set it to **None** to regain free manual control.
- **"Custom" appears in Scenario box:** This is normal. If you edit any physics parameter manually, the app switches the label to "Custom" to indicate the settings no longer match a preset.

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10) Release Notes

v1.3.0

- **Feature:** Added **Ground Station 2**. The application now supports two independent ground stations, each with its own coordinates, elevation mask, and access calculations.
- **UI:** New "Ground Station 2" panel (disabled by default). GS1 shown as white triangle, GS2 as green triangle on both 2D and 3D views.
- **UI:** Access intervals table now includes a "Station" column and merges both stations' intervals chronologically.
- **UI:** Separate footer badges for GS1 and GS2 real-time access status.
- **Export:** CSV now labels intervals as `ground-station-1` or `ground-station-2`.
- **Code Quality:** Removed stray Evernote/Skitch CSS artifacts (176 lines) and browser-injected comment from file.
- **Documentation:** Updated User Manual with new Ground Station Panels section (5.3).

v1.2.0

- **Feature:** Added **TLE Import** to parse Two-Line Element sets and populate orbital elements.
- **UI:** Collapsible "Import from TLE" section in each satellite panel with epoch sync option.
- **Keyboard:** Added `T` (Sat 1) and `Shift+T` (Sat 2) shortcuts for TLE import.
- **Documentation:** Updated User Manual with new TLE Import section (5.6).

v1.1.2

- **Feature:** Added **Keyboard Shortcuts** for improved workflow (Space, R, arrow keys, V, I, 0/1/2, E).
- **Documentation:** Updated User Manual with new Keyboard Shortcuts section (5.5).

v1.1.1

- **Fix:** Corrected negative longitude and angle wrapping functions.
- **Fix:** Added numerical tolerance for 90° elevation mask edge case.
- **Fix:** Removed dead UI references and unified terminator toggle handling.
- **Code Quality:** Renamed internal True Anomaly variables from `M` to `nu` for semantic correctness.

v1.1.0

- **Feature:** Added **Scenario Presets**. Users can now instantly load LEO, SSO, GEO, and Molniya configurations via a dropdown menu.
- **UI:** Added logic to auto-detect manual edits and switch scenario state to "Custom."
- **Core:** Validated J2 propagator against standard SSO parameters.
- **Documentation:** Updated User Manual to reflect the new workflow.